MULTIPLYING A TALL MATRIX BY THE ADJUGATE OF ITS TOP BLOCK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The adjugate of a square matrix multiplies to the determinant. Used at the first line of the counted proof (line 2953). Gives A_1 B = B A_1 = f, which settles the top block of the product.
- Row expansion of a determinant. Used at the end of the counted proof (lines 2959-2960). Identifies the entry of the product with a signed maximal minor of the original matrix.

Lemma 0.1. *Let R be a ring. Let $n \geq m$. Let $A = (a_{ij})$ be an $n \times m$ matrix with coefficients in R, written in block form as*

$$A = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

where A_1 has size $m \times m$. Let B be the adjugate (transpose of cofactor) matrix to A_1. Then

$$AB = \begin{pmatrix} f1_{m \times m} \\ C \end{pmatrix}$$

where $f = \det(A_1)$ and c_{ij} is (up to sign) the determinant of the $m \times m$ minor of A corresponding to the rows $1, \ldots, \hat{j}, \ldots, m, i$.

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Proof. Since the adjugate has the property $A_1B = BA_1 = f$ the first block of the expression for AB is correct. Note that

$$c_{ij} = \sum\nolimits_k a_{ik}b_{kj} = \sum (-1)^{j+k} a_{ik} \det(A_1^{jk})$$

where A_1^{ij} means A_1 with the jth row and kth column removed. This last expression is the row expansion of the determinant of the matrix in the statement of the lemma. $\square$